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Program : Computer Engineering Program Date : **1 March 2026**
Examiners : Dr. Tarek salem
Eng :Habiba,Basant , Eman Duration **2 hours**
Course/Code : Database Management Systems / CSE112

Part 1: Quick Summary**1) What is main properties of database ?**

three main properties:

1. A database represents part of the real world, called the **mini-world** or **Universe of Discourse (UoD)**.
Any changes in the real world are reflected in the database.
2. A database is an organized and meaningful collection of data.
It is not just random data collected together.
3. A database is created for a specific purpose and for a specific group of users.

2) What are the main functions of a DBMS?

A **DBMS** is a software system that allows users to create and manage databases.

It supports:

Defining the database (data types, structures, and constraints), where the description is stored as **meta-data** in a catalog.

Constructing the database by storing data on a storage medium.

Manipulating the database through queries, updates, and reports.

Sharing the database among multiple users and applications simultaneously.

3)How does an Application Program access the database?

Answer:

The **Application Program** does not deal with the database directly.
It sends requests to the **DBMS (Database Management System)**, which then retrieves or modifies the data and returns the result.

Application → DBMS → Database

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3) What is a Query?

A Query is a request sent from the application to retrieve specific data from the database.

It usually: **Reads data - Does not modify it** **Query = Data Retrieval**

4) What is a Transaction?

A Transaction is a complete operation that may: **Read data - Write or modify data**

It must be executed fully or not executed at all to keep the database consistent.

Transaction = Read + Write Operations

6) Why does the database system need Protection?

Because queries and transactions access important data, the system must be protected.

Protection includes:

System Protection:

Protects against system failures and ensures data recovery.

Security Protection:

Prevents unauthorized access and protects sensitive data.

7) Why is Maintaining the Database System important?

Answer:

Because requirements change over time.

Maintenance allows the system to: Adapt to new needs - Improve performance - Enhance security

Concept Flow (Connected Idea)

Application → sends Query or Transaction → DBMS processes → Protection ensures safety → Maintenance ensures long-term evolution.

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Example: UNIVERSITY Database**Description**

The UNIVERSITY database stores information about students, courses, and grades in a university environment.

Main Records

The database includes the following entities:

- STUDENT
- COURSE
- SECTION
- GRADE_REPORT
- PREREQUISITE

1 .Defining Record Structure

Each record must have a defined structure by specifying the data type of each attribute, such as:

String (e.g., name, course title)

Integer (e.g., ID, section number)

2.Database Construction

To build the database

3.Define the relationships among the records (e.g., a student enrolls in a course, a course has prerequisites).

4.Perform data manipulation operations such as:

Querying (retrieving data)

Updating (modifying data)

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STUDENT

Name	Student_number	Class	Major
Smith	17	1	CS
Brown	8	2	CS

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	07	King
92	CS1310	Fall	07	Anderson
102	CS3320	Spring	08	Knuth
112	MATH2410	Fall	08	Chang
119	CS1310	Fall	08	Anderson
135	CS3380	Fall	08	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

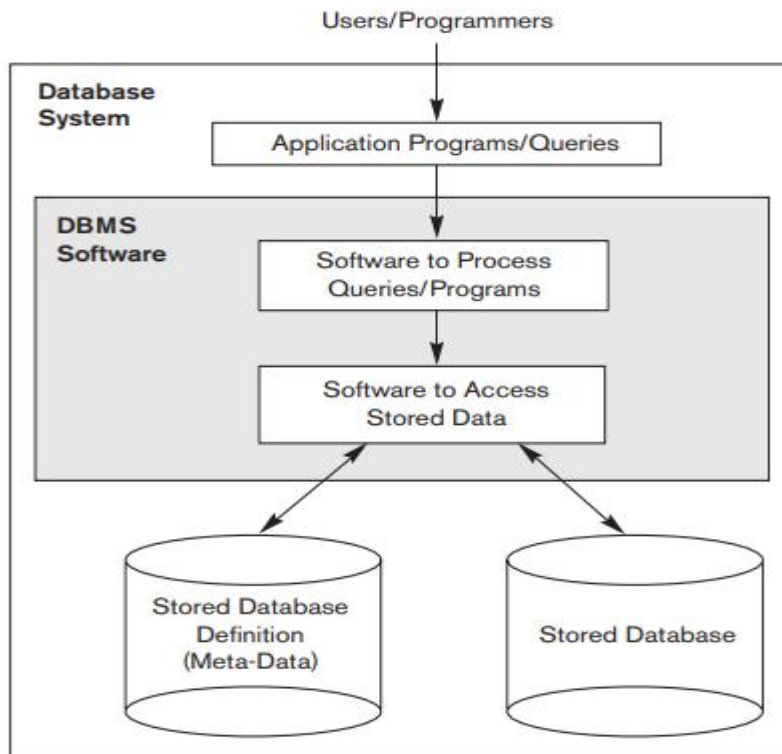
Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

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Database System Architecture

This figure shows how users interact with the database system.

Users or programmers send **application programs or queries** to the DBMS. The **DBMS software** processes these queries and accesses the stored data.

The system depends on two main components:

Stored Database (actual data).

Stored Database Definition (Meta-Data), which describes the structure of the database.

The DBMS acts as an intermediary between users and the database to ensure proper processing and data access.

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Differentiate between Actors on the Scene and Workers Behind the Scene in a Database System?

In a database system, people involved are classified into two main categories:

1- Actors on the Scene

They directly interact with the database system.

They include:

- **Database Administrators (DBA)** → Manage and control the database.
- **Database Designers** → Design the database structure.
- **End Users** → Use the system to retrieve or update data.
- **Application Programmers** → Develop programs that access the database.

They work directly with the database system.

2- Workers Behind the Scene

They support the database system but do not directly use it.

They include:

- DBMS system designers and implementers.
- Tool developers.
- Operators and maintenance staff.

They ensure the system runs efficiently and correctly.

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Part 2: Questions

A) Multiple Choice (MCQ)

1) Data is defined as:

- a) Processed information
- b) Raw facts and figures
- c) Organized tables
- d) Software programs

2) A database represents:

- a) Random data
- b) A programming language
- c) A mini-world (real-world part)
- d) Hardware storage

3) The main purpose of a DBMS is to:

- a) Design websites
- b) Manage and control databases
- c) Replace operating systems
- d) Write algorithms

4) Data redundancy means:

- a) Data stored once
- b) Duplicate copies of data
- c) Encrypted data
- d) Sorted data

5) Which of the following is a modern database application?

- a) Student transcript system
- b) Banking system
- c) Multimedia database
- d) Inventory system

6) SQL is considered:

- a) Procedural language
- b) Object-oriented language
- c) Declarative language
- d) Machine language

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7) Which of the following is NOT a programming language?

- a) Java
- b) C++
- c) Python
- d) **SQL**

8) A transaction may:

- a) Only read data
- b) Only delete data
- c) **Read and write data**
- d) Do nothing

9) Meta-data is:

- a) Raw student data
- b) **Data about data**
- c) Video files
- d) Backup files

10) The physical level of database architecture describes:

- a) User views
- b) Table relationships
- c) **How data is stored on disk**
- d) SQL queries

11) Which of the following is a main property of a database?

- a) It stores random data
- b) **It represents part of the real world**
- c) It works without users
- d) It does not need maintenance

12) Which problem occurs when the same data is stored in multiple files?

- a) Data isolation
- b) **Data redundancy**
- c) Data security
- d) Data abstraction

13) Which of the following is considered an Actor on the Scene?

- a) Tool developer
- b) System operator
- c) **Database Administrator (DBA)**
- d) Hardware engineer

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14) Which level of database architecture hides some details for security?

- a) Physical level
- b) Logical level
- c) View level**
- d) Storage level

15) NOSQL databases are mainly designed to:

- a) Replace SQL completely
- b) Handle large-scale unstructured data**
- c) Store only numeric data
- d) Work without servers

16) A transaction must follow which principle?

- a) Partial execution
- b) Execute completely or not at all**
- c) Only read data
- d) Only update data

17) Which of the following is NOT a function of a DBMS?

- a) Defining database structure
- b) Managing data security
- c) Designing computer hardware**
- d) Controlling data access

18) In a database system, meta-data is stored in the:

- a) Application program
- b) System catalog**
- c) User interface
- d) Operating system

19) Which of the following best describes the Logical Level of database architecture?

- a) It shows how data is physically stored.
- b) It describes the overall structure of the database.**
- c) It hides information from users completely.
- d) It represents hardware components.

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20) What is the main advantage of using a database instead of a file system?

- a) Increased data duplication
- b) Reduced data consistency
- c) **Better organization and centralized control**
- d) No need for security mechanisms

B) True or False

1. A database is just a random collection of data. **False**
2. Metadata describes the structure of the database. **True**
3. Programming languages focus on what data we want, not how to get it. **False**
4. SQL is mainly used to create and manage data inside databases. **True**
5. The DBMS acts as an intermediary between users and the database. **True**
6. Actors on the Scene directly interact with the database system. **True**
7. A database is created for a specific purpose and group of users. **True**
8. SQL tells the system how to search step by step. **False**
9. Data warehouses are used for historical data analysis. **True**
10. Workers Behind the Scene directly query the database. **False**

C) Complete the Following

1. A database is a collection of ____ **Related** ____ data.
2. The three levels of database architecture are:
____ **Physical, Logical, View** ____
3. A Query is mainly used for ____ **Retrieving** ____ data.

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4. Two types of database protection are:
___System___ protection and ___Security___ protection.
- 5 .A DBMS helps in ___defining___, ___constructing___, and manipulating databases.
6. Two examples of programming languages are ___java___ and ___c++___.
7. In real applications, ___programming___ language handles logic, while ___Sql___ manages data.
8. A database represents a part of the real world called the ___mini-world (UoD)___.
9. The description of the database structure is stored as ___meta data___ in the system catalog.
10. Actors on the Scene include the ___ Database Administrator (DBA)___ and application programmers.
-

D)Design a simple database for a Hospital System.

1. Write four main entities.

Entities:

- Patient
- Doctor
- Appointment
- Treatment

2. Give two examples of relationships between them.

Relationships:

- Patient books appointment with doctor
- Doctor gives treatment to patient

3. Mention two possible data types used.

- String
- Integer